

Supplementary Information

A deep convolutional neural network for regression analysis of X-ray diffraction computed tomography data

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Supplementary Note 1. Glossary of terms

Fully connected layer	Each neuron in a fully connected layer is connected to every neuron in the previous layer. It can create n^2 trainable weights when two fully connected layers with n nodes are connected together.
Dense layer	Another name for fully connected layer.
Kernel	The scrolling window in the max-pooling layers and convolutional layers that operates only a part of the input image at a time.
Number of filters	The number of kernels in one convolutional layer
Max-pooling layer	A max-pooling layer usually has only one kernel and takes only the maximum value in a part of the image. The image sizes are decreased after the max-pooling layers.

Convolutional layer	A feature map of the input image is yielded by the feature kernels in convolutional layers. The output channels are equal to the number of kernels used in the convolutional layer.
Dropout layer	A layer normally used after fully connected layers to reduce overfitting in artificial neural networks. After applying a dropout layer, a proportion of weights in the fully connected layer are omitted while training. The omitted weights are randomly selected, and will be different in each training epoch.
Flatten layer	A layer that converts multidimensional outputs to 1D array. The flatten layer does not have any trainable weights.

Supplementary Note 2. Simulated single-phase XRD-CT dataset

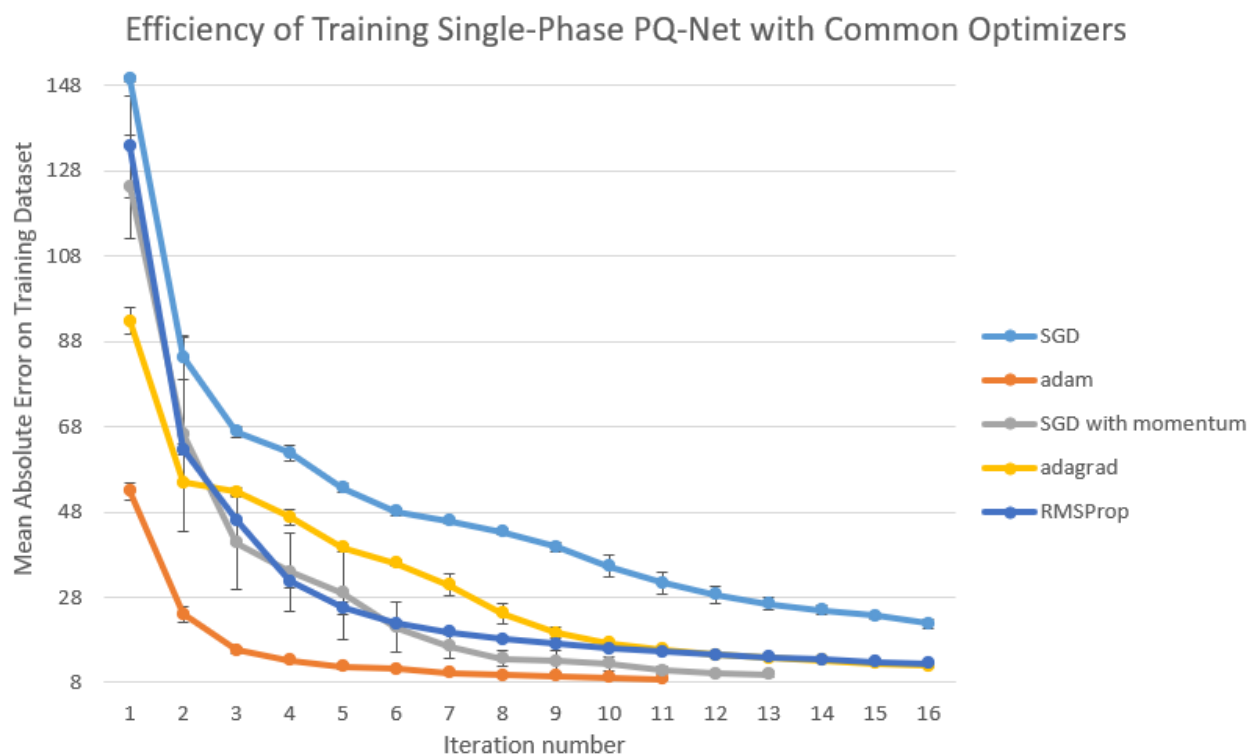
Supplementary Note 2.1. Differences in Models’ Performance with CPU and GPU

Validation split: 10%
Batch size: 32
Loss function: Mean absolute error (MAE)

Supplementary Table 1.

Training library size		Training time per epoch (s)	Predicting time on 1k patterns (s)
1,000	CPU	27	5.54
	GPU	2	0.61
10,000	CPU	279	5.54
	GPU	14	0.61
50,000	CPU	1396	5.54
	GPU	70	0.61

Supplementary Note 2.2. Differences in Models’ Performance using various optimisation algorithms



Supplementary Figure 1. Comparison among the performance of common optimizers. MAE is chosen as the loss function. The training was performed on single-phase PQ-Net architectures with 10,000 training dataset with learning rate as 0.0005. Not all optimizers did 16 epochs because they have got enough accuracy, and the validation loss did not improve anymore.

Supplementary Note 2.3. Accuracy Differences with Different Parameter Settings in the Single-Phase PQ-Net

Training dataset size: 20,000

Validation split : 10%

Optimizer = Adam

Conclusion : increasing the filter numbers does not necessarily mean a better performance, but it can easily lead to local minimas especially when the training dataset is small. By considering the balance of model's performance and the training time, we chose to use the parameter settings shown in Figure 1.

Supplementary Table 2. The MAEs of Validation Dataset with Different Layer Sizes in Single-Phase PQ-Net

	With 0.5x filter numbers	With original convolutional layer size	With 2x filter numbers
With 0.5 x nodes in fully connected layers	5.3	5.49	6.08

With original fully connected layer size	3.47	3.28	4.07
With 2 x nodes in fully connected layers	3.21	3.23	3.01

Supplementary Note 2.4. Impact of Library Size

Library: Simulated Ni catalyst XRD-CT dataset

Library sizes: 1,000, 2,000, 5,000, 10,000, 20,000, 50,000, 100,000 and 250,000

Learning rate : 0.0005

Loss function: Mean absolute error (MAE)

Q range: 0-10 Å⁻¹

Pattern Size: 2000 bins

Batch size: 32

Training time: 36 ms/step

Ni scale factor range: random sampling between 0-4 × 10⁻⁹

Ni lattice parameter range: 3.55 ± 0.025 Å

Ni crystallite size range: 2-25 nm

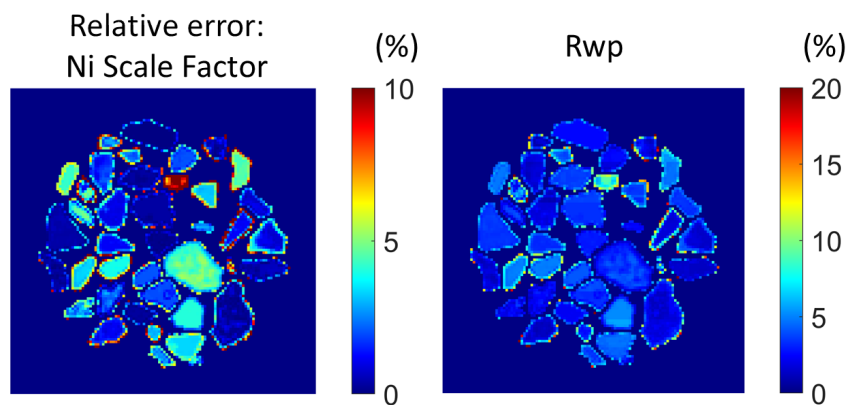
Early stopping: 3

Supplementary Table 3.

Library size	Epochs	MAE	Val MAE
1K	21	17.3640	28.2726
2K	24	7.8272	7.7344
5K	17	6.0237	7.1791
10K	13	5.8395	8.1244
20K	12	4.0839	3.8190
50K	11	2.8442	3.0839
100K	25	1.2830	1.4234
250K	13	1.2046	1.6934

Supplementary Note 2.5. Relative Error and Rwp

The error maps for the PQ-Net trained with 100K patterns using the simulated single-phase XRD-CT dataset are shown in Supplementary Figure 2.



Supplementary Figure 2. Relative error map for the Ni scale factor by comparing the results obtained with the PQ-Net and the ground truth and Rwp map as calculated with TOPAS software v7.

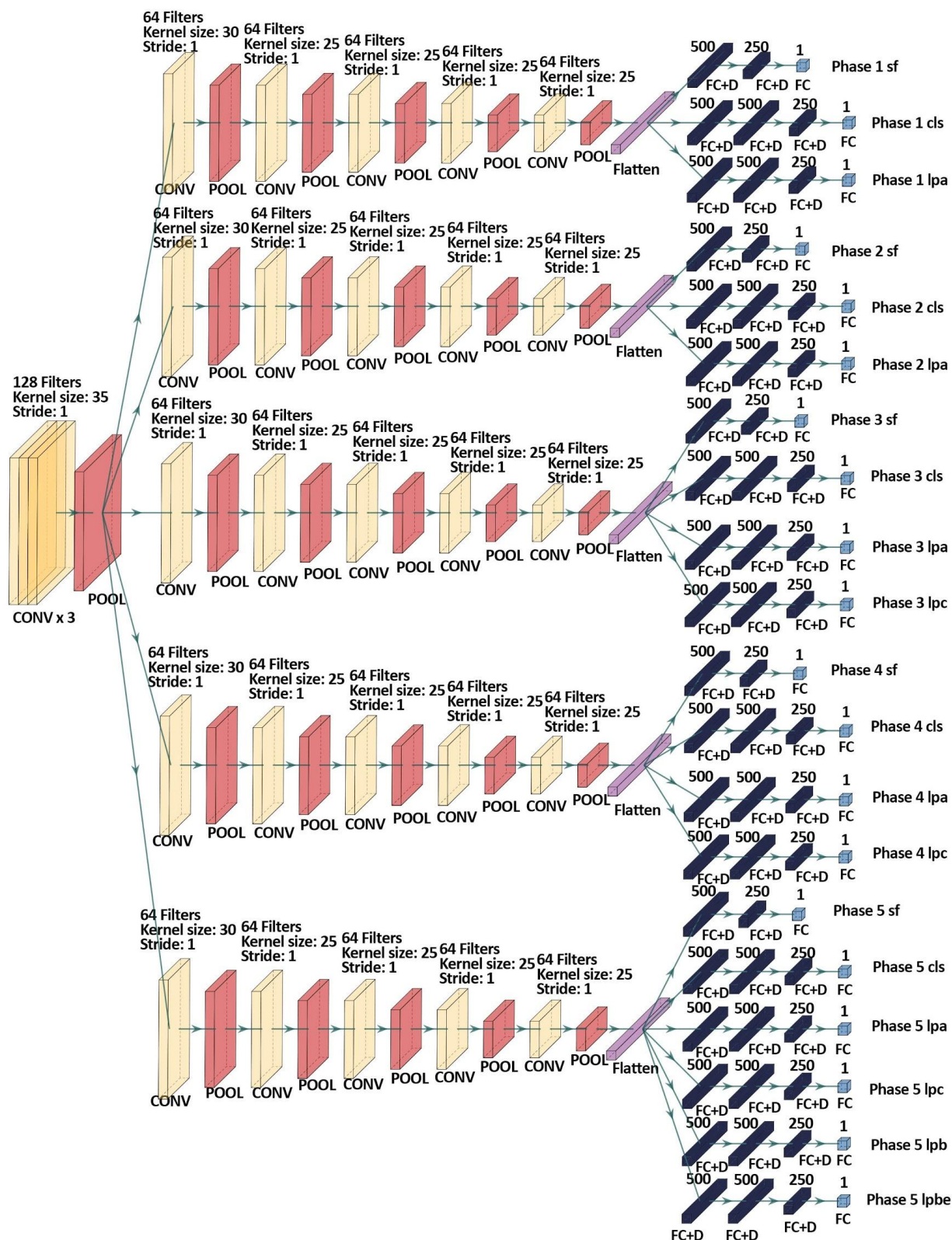
Supplementary Note 2.6. Phase parameters for particles-of-interest

Supplementary Table 4.

Ni		Lattice Parameter (Å)	Crystallite Size (nm)
Particle 1	Ground Truth	3.544	21.6
	PQ-Net	3.543	23.4
Particle 2	Ground Truth	3.539	7.6
	PQ-Net	3.539	7.6
Particle 3	Ground Truth	3.558	4.5
	PQ-Net	3.558	4.6

Supplementary Note 3. Multi-phase PQ-Net and Deep Ensemble

The five-phase PQ-Net architecture is presented in Supplementary Figure 3.



Supplementary Figure 3. Regression CNN architecture for POX catalyst experimental system with five phases. CONV represents 1-D convolutional layers, POOL represents max-pooling layers, FC represents fully connected layers, and FC+D represents a fully connected layer and a dropout layer. The CNN splits after the first max-pooling layer and forms five sub-CNNs corresponding to the five phases in the experimental system. As shown in the schematic, each parameter has its own route after flattening layers. Scales of fully connected layers are shown above them. There is also a 10% dropout layer after each hidden fully connected layer. The number of filters, kernel sizes and the stride of convolutional layers are also given. All max-pooling layers have stride equals 2.

Supplementary Note 3.1. Parameters

NiO scale factor range: random sampling between $0-1 \times 10^{-9}$

NiO lattice parameter a range: $4.137 \pm 0.02 \text{ \AA}$

NiO crystallite size range: 2-50 nm

CeO₂ scale factor range: random sampling between $0-1 \times 10^{-10}$

CeO₂ lattice parameter a range: $5.350 \pm 0.05 \text{ \AA}$

CeO₂ crystallite size range: 2-50 nm

ZrO₂ scale factor range: random sampling between $0-1 \times 10^{-9}$

ZrO₂ lattice parameter a range: $3.589 \pm 0.02 \text{ \AA}$

ZrO₂ lattice parameter c range: $5.230 \pm 0.02 \text{ \AA}$

ZrO₂ crystallite size range: 2-50 nm

PdO scale factor range: random sampling between $0-1 \times 10^{-9}$

PdO lattice parameter a range: $3.010 \pm 0.02 \text{ \AA}$

PdO lattice parameter c range: $5.350 \pm 0.02 \text{ \AA}$

PdO crystallite size range: 2-50 nm

Al₂O₃ scale factor range: random sampling between $0-1 \times 10^{-11}$

Al₂O₃ lattice parameter a range: $11.965 \pm 0.02 \text{ \AA}$

Al₂O₃ lattice parameter b range: $11.133 \pm 0.02 \text{ \AA}$

Al₂O₃ lattice parameter c range: $11.071 \pm 0.02 \text{ \AA}$

Al₂O₃ lattice parameter beta range: $103.246 \pm 1^\circ$

Al₂O₃ crystallite size range: 2-50 nm

Supplementary Note 3.2. Single Model

Library: Simulated Ni-Pd/CeO₂-ZrO₂/Al₂O₃ catalyst XRD-CT dataset

Library sizes: 1,000, 2,000, 5,000, 10,000, 20,000, 50,000, 100,000 and 250,000

Learning rate : 0.0005

Loss function: Mean absolute error (MAE)

Q range: $1.8-8 \text{ \AA}^{-1}$

Pattern Size: 740 bins

Batch size: 32

Training time: 53 ms/step

Early stopping: 5

Added features: Poisson noise and 2nd degree Chebyshev polynomial

Supplementary Table 5.

Library size	Epochs	MAE	Val MAE
1K	81	130.1829	243.5705
2K	78	76.31293	164.31412
5K	43	88.65284	135.48003
10K	53	88.29119	125.9081
20K	47	84.54045	115.319
50K	31	71.31851	99.78559
100K	38	85.49037	96.30725
250K	34	71.94207	80.57399

Supplementary Note 3.3. Deep Ensemble

Library: Simulated Ni-Pd/CeO₂-ZrO₂/Al₂O₃ catalyst XRD-CT dataset

Library sizes: 10,000, 30,000, 50,000 and 100,000

Learning rate : 0.0005

Loss function: Mean absolute error (MAE)

Pattern Size: 740 bins

Batch size: 32

Training time: 82 ms/step

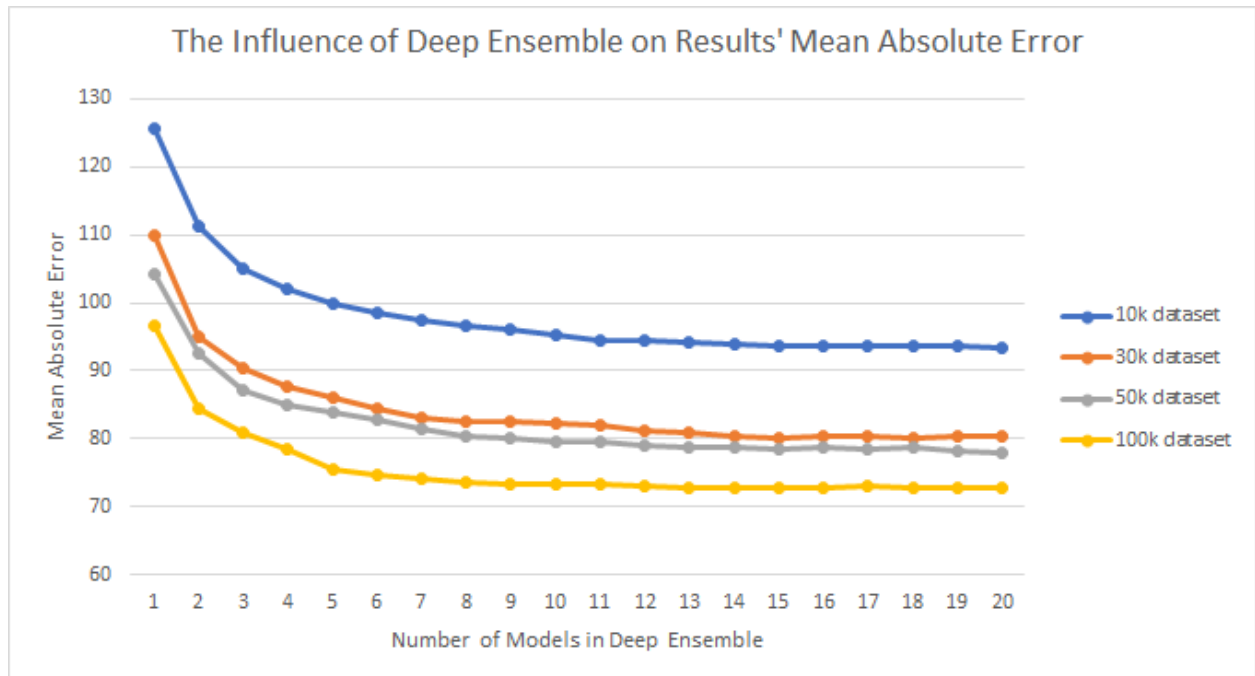
Added features: Poisson noise and 2nd degree Chebyshev polynomial

Early stopping: 2

The results presented in Supplementary Table 6 corresponds to the MAE of the validation datasets. Since the uncertainties for single models are very large, when DE sizes are less than 5, the MAEs shown in the table are the average MAEs among 5 same-sized DE models (All individual models used in DE are unique).

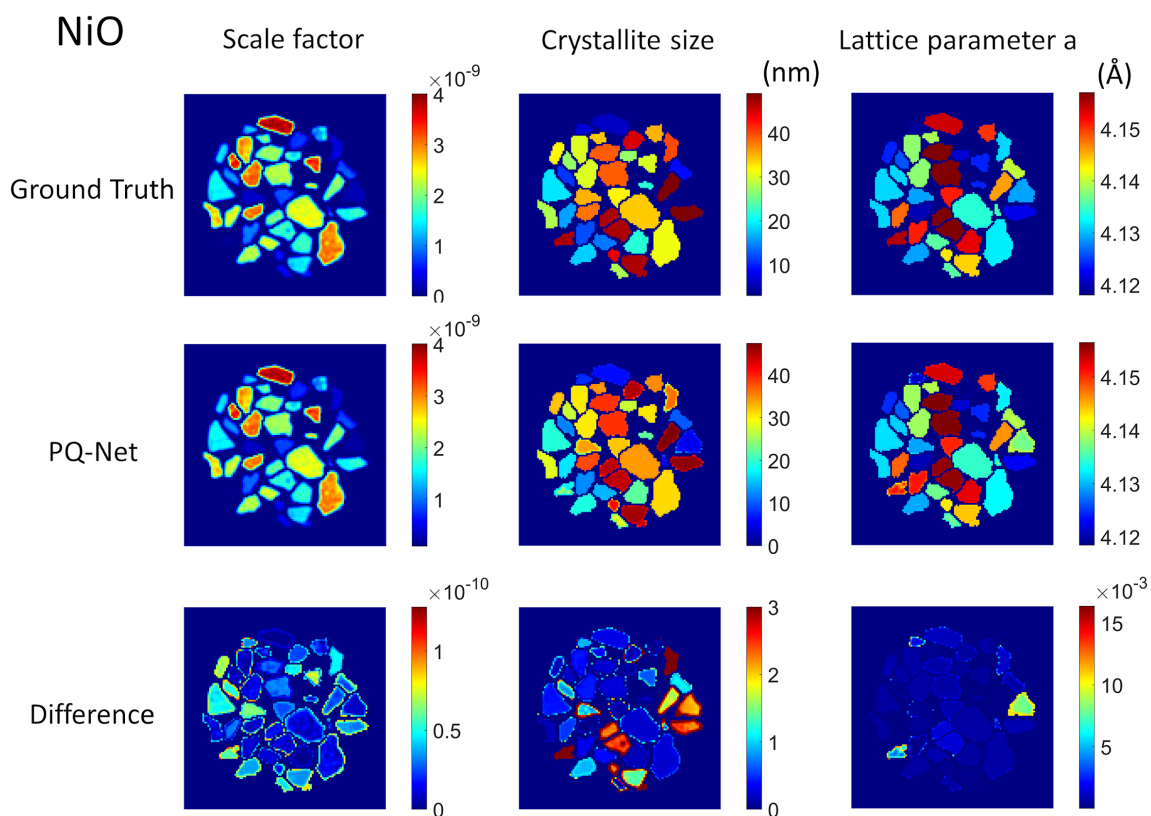
Supplementary Table 6.

DE size	10k	30k	50k	100k
1	125.6507	109.8903	104.3413	96.65125
2	111.2758	94.92358	92.7044	84.3195
3	105.0534	90.44386	87.21929	80.9583
4	102.0792	87.71534	84.97091	78.45387
5	99.80675	85.97904	83.79546	75.51462
6	98.45306	84.46986	82.72748	74.77451
7	97.31806	83.21263	81.50329	74.18023
8	96.72295	82.54938	80.52089	73.74579
9	95.99856	82.46798	80.21869	73.352
10	95.21096	82.20558	79.7026	73.21912
11	94.33005	81.91061	79.63169	73.30816
12	94.40285	81.23354	79.13793	73.00534
13	94.14324	80.85489	78.88585	72.87525
14	93.93362	80.47411	78.78531	72.73217
15	93.78373	80.24465	78.59158	72.91953
16	93.69253	80.26007	78.74517	72.85157
17	93.60937	80.48855	78.57577	73.02039
18	93.61759	80.15814	78.71829	72.89617
19	93.52919	80.36047	78.27454	72.82898
20	93.35656	80.36112	78.04849	72.72646

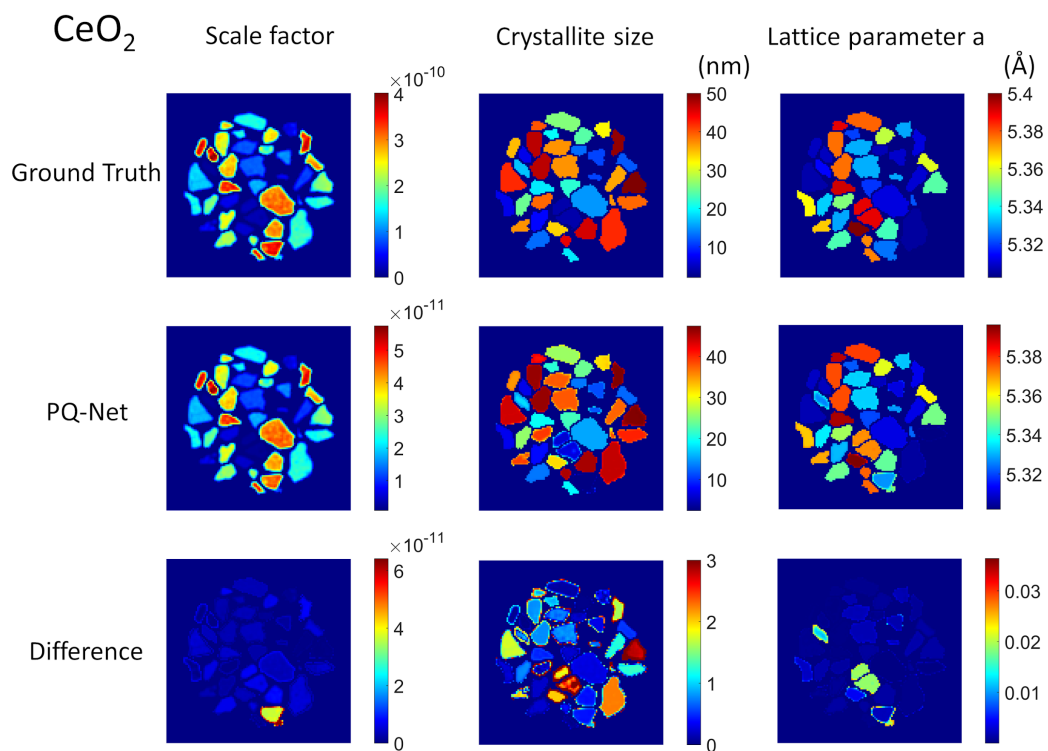


Supplementary Figure 4. The influence of the deep ensemble sizes on models' performance. The points in the chart represent the sum of 20 outputs' MAEs in the simulated validation dataset. All curves tend to flatten out as the number of models in deep ensemble increases; and the larger the training dataset is, the faster it flattens out. Since the uncertainties of individual models are very large, when the deep ensemble size is less than 5, the MAE shown in the figure is the average MAE among five same-sized deep ensemble models (All individual models are unique).

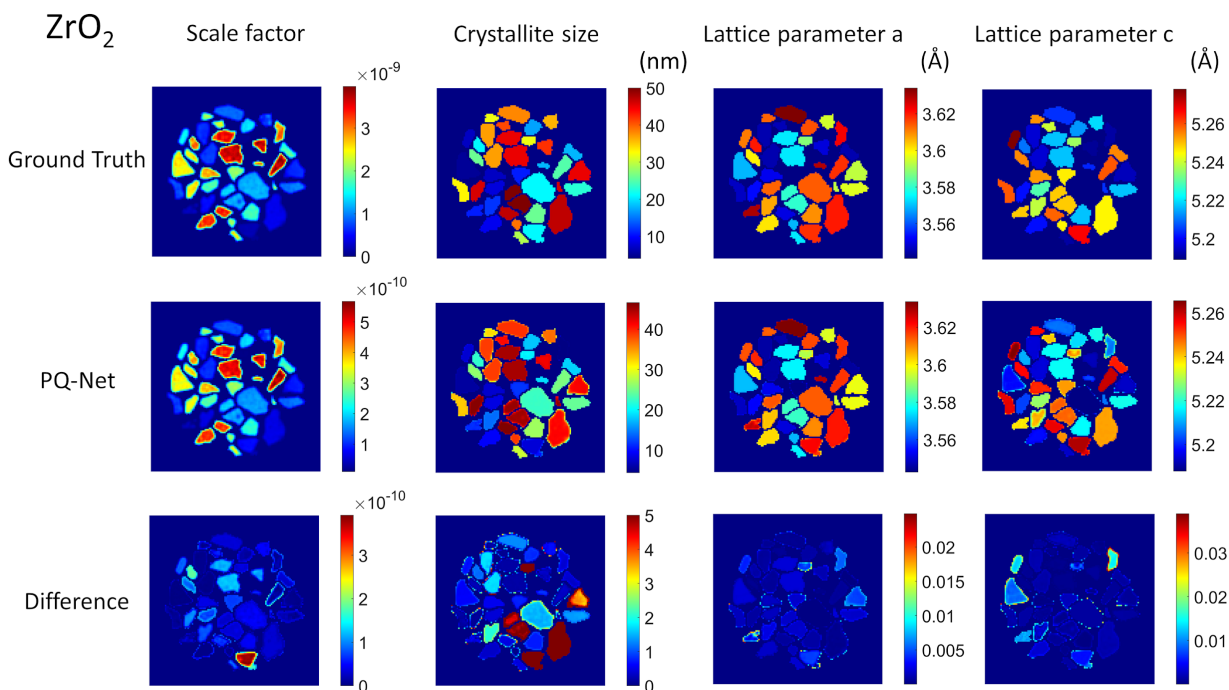
Supplementary Note 3.4. Phase parameter maps



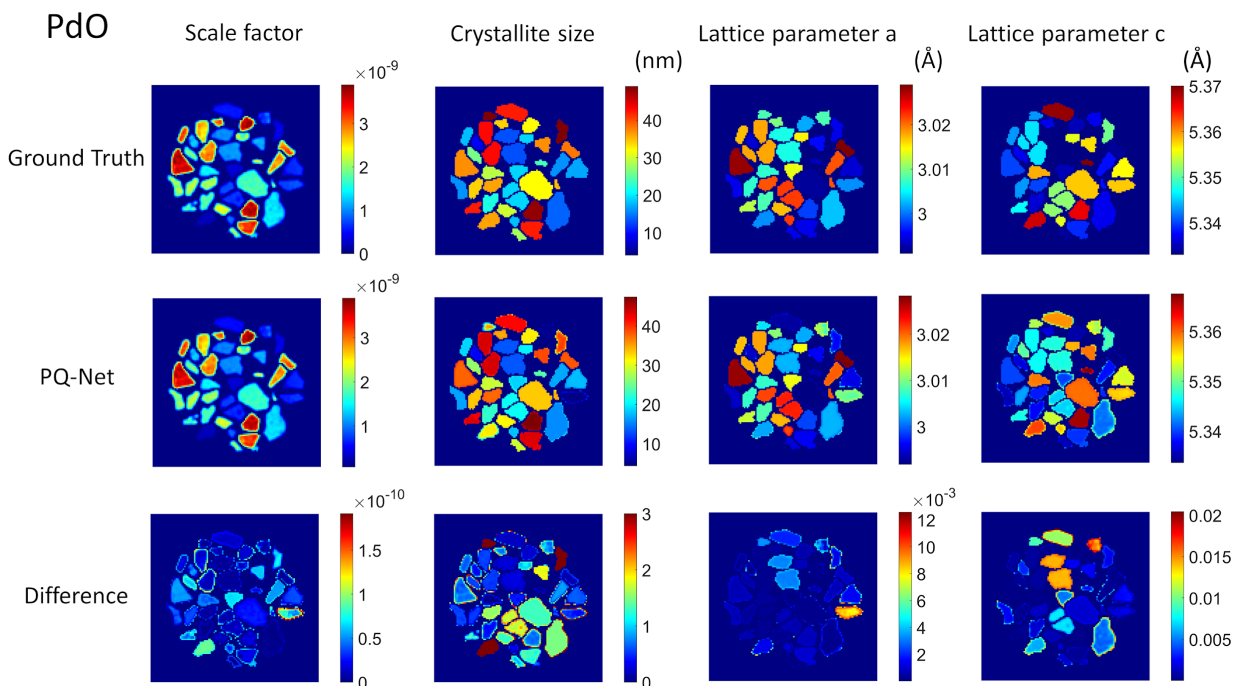
Supplementary Figure 5. Comparison of the scale factor, lattice parameter and crystallite size maps obtained with the PQ-Net (10 model deep ensemble trained with 100K patterns) for the NiO phase and the ground truth values.



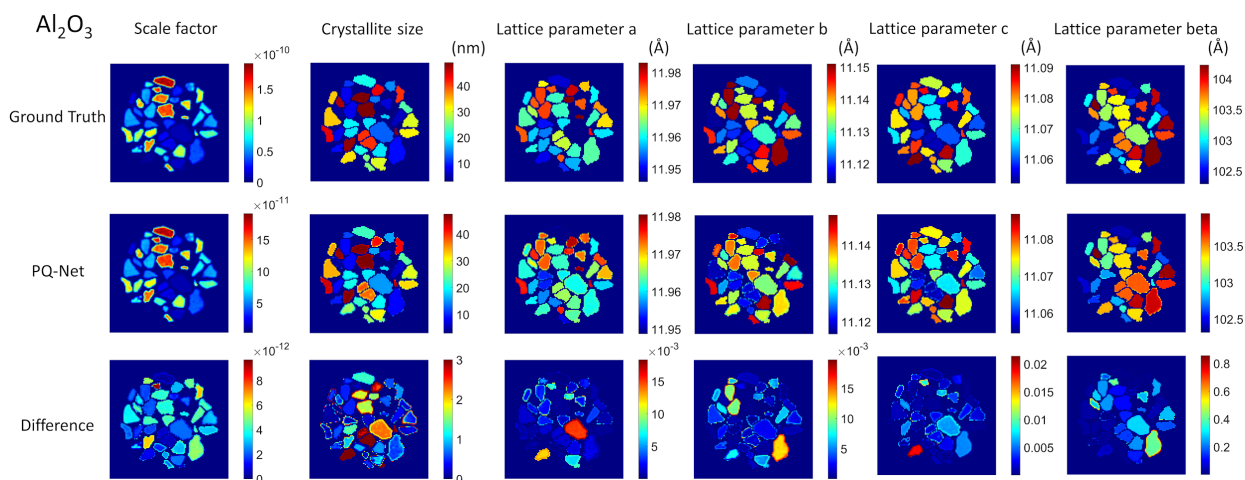
Supplementary Figure 6. Comparison of the scale factor, lattice parameter and crystallite size maps obtained with the PQ-Net (10 model deep ensemble trained with 100K patterns) for the CeO₂ phase and the ground truth values.



Supplementary Figure 7. Comparison of the scale factor, lattice parameter and crystallite size maps obtained with the PQ-Net (10 model deep ensemble trained with 100K patterns) for the ZrO_2 phase and the ground truth values.

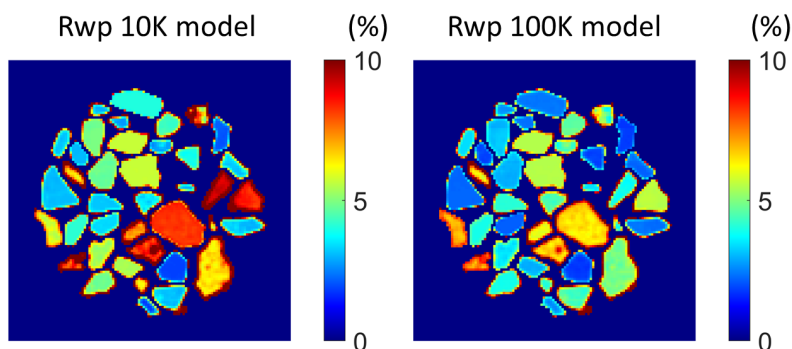


Supplementary Figure 8. Comparison of the scale factor, lattice parameter and crystallite size maps obtained with the PQ-Net (10 model deep ensemble trained with 100K patterns) for the PdO phase and the ground truth values.



Supplementary Figure 9. Comparison of the scale factor, lattice parameter and crystallite size maps obtained with the PQ-Net (10 model deep ensemble trained with 100K patterns) for the Al_2O_3 phase and the ground truth values.

The error maps for the deep ensemble PQ-Net consisting of 10 models and trained with 10K and 100K patterns using the simulated multi-phase XRD-CT dataset are shown in Supplementary Figure 10.



Supplementary Figure 10. Comparison of Rwp maps obtained with a 10 model deep ensemble PQ-Net trained with 10K and 100K patterns respectively.

Supplementary Note 3.5. Phase parameters for particles-of-interest

Supplementary Table 7.

NiO		Lattice Parameter a (Å)	Crystallite Size (nm)
Particle 1	Ground Truth	4.153	22
	PQ-Net	4.1525	22.2
Particle 2	Ground Truth	4.144	47
	PQ-Net	4.1441	45.9
Particle 3	Ground Truth	4.132	31
	PQ-Net	4.1328	31.2

Supplementary Table 8.

CeO2		Lattice Parameter a (Å)	Crystallite Size (nm)
Particle 1	Ground Truth	5.345	46
	PQ-Net	5.346	45.6
Particle 2	Ground Truth	5.325	2
	PQ-Net	5.329	2.4

Particle 3	Ground Truth	5.305	42
	PQ-Net	5.305	43.1

Supplementary Table 9.

ZrO ₂		Lattice Parameter a (Å)	Lattice Parameter c (Å)	Crystallite Size (nm)
Particle 1	Ground Truth	3.609	5.217	26
	PQ-Net	3.610	5.217	26.8
Particle 2	Ground Truth	3.618	5.267	21
	PQ-Net	3.624	5.263	8.4
Particle 3	Ground Truth	3.62	5.245	47
	PQ-Net	3.621	5.245	42.9

Supplementary Table 10.

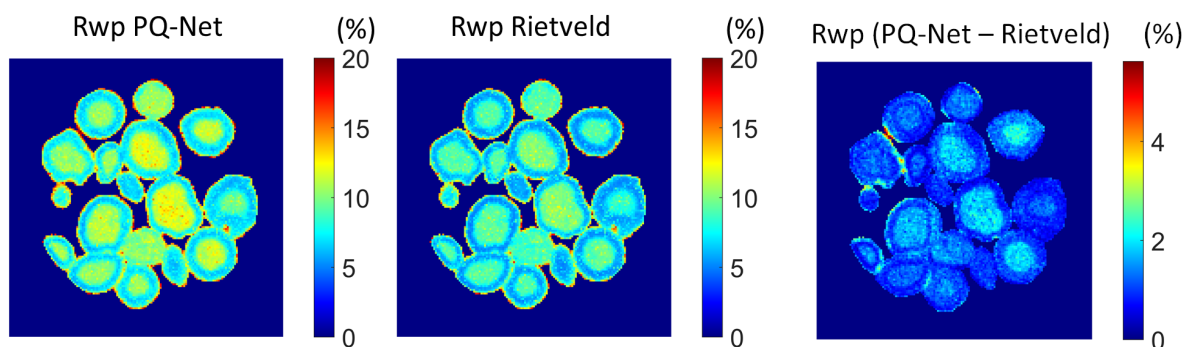
PdO		Lattice Parameter a (Å)	Lattice Parameter c (Å)	Crystallite Size (nm)
Particle 1	Ground Truth	2.998	5.366	48
	PQ-Net	2.999	5.366	47.4
Particle 2	Ground Truth	2.995	5.339	42
	PQ-Net	2.996	5.339	42.5
Particle 3	Ground Truth	3.003	5.337	14
	PQ-Net	3.003	5.338	15.2

Supplementary Table 11.

Al ₂ O ₃		Lattice Parameter a (Å)	Lattice Parameter b (Å)	Lattice Parameter c (Å)	Lattice Parameter beta (°)	Crystallite Size (nm)
Particle	Ground	11.962	11.146	11.089	104.068	19

1	Truth					
	PQ-Net	11.964	11.146	11.087	103.838	19.8
Particle 2	Ground Truth	11.957	11.14	11.072	103.501	30
	PQ-Net	11.963	11.137	11.070	103.382	31.8
Particle 3	Ground Truth	11.963	11.15	11.068	104.219	9
	PQ-Net	11.959	11.139	11.074	103.870	9.3

Supplementary Note 4. Experimental multi-phase XRD-CT dataset



Supplementary Figure 11. Comparison of Rwp maps obtained with a 10 model deep ensemble PQ-Net trained with 100K and with the Rietveld method.

Supplementary Note 5. Summary of test datasets

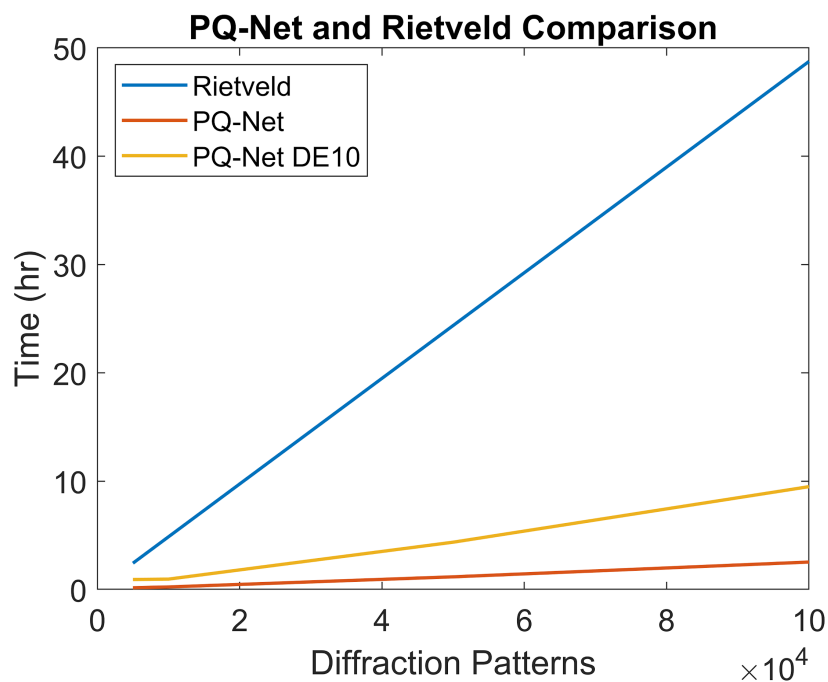
Supplementary Table 12. Size of the three test datasets used in this work

Test Dataset	Ni Simulated	Simulated Ni-Pd/CeO ₂ -ZrO ₂ /Al ₂ O ₃	Experimental Ni-Pd/CeO ₂ -ZrO ₂ /Al ₂ O ₃
Size	14400 (120 x 120)	14400 (120 x 120)	22801 (151 x 151) for the PQ-Net, 9027 for the Rietveld after cropping

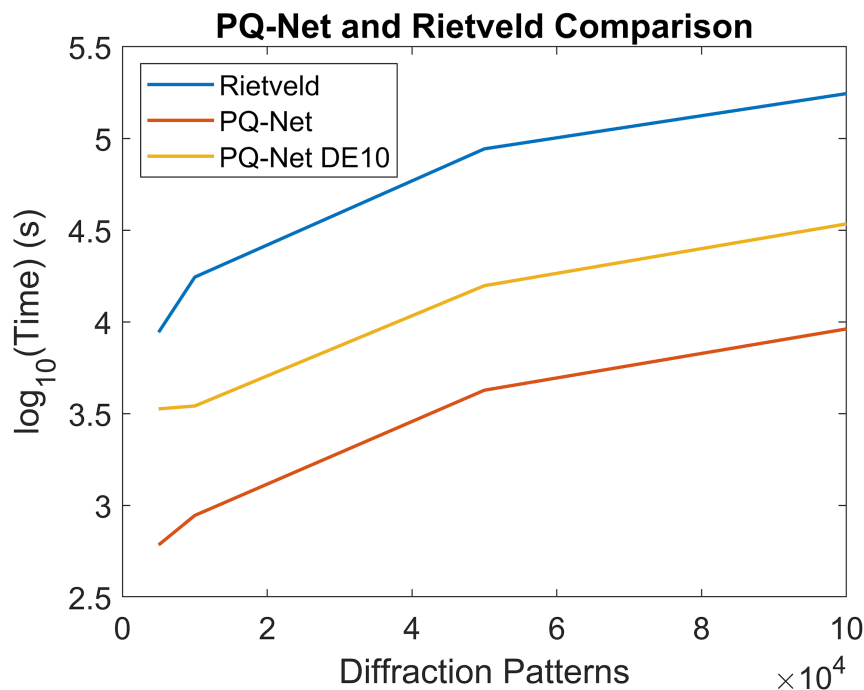
Supplementary Note 6. Timing comparison between PQ-Net and Rietveld for the multi-phase dataset

Supplementary Table 13.

Library size	Library creation time (s)	Training time per epoch (s)	Total epochs to converge	Total training time (s)	Prediction time (s)	PQ-Net total time (hr)	Time taken by Rietveld method (hr)
5,000	293	9	34	313	2.22	0.17	2.4
10,000	586	17	17	292	3.78	0.25	4.9
50,000	2930	86	15	1293	19.71	1.2	24.4
100,000	5860	185	18	3312	41.11	2.5	48.8



Supplementary Figure 12. Performance comparison between the Rietveld method, single-model PQ-Net and PQ-Net Deep Ensemble with 10 models.



Supplementary Figure 13. Performance comparison between the Rietveld method, single-model PQ-Net and PQ-Net Deep Ensemble with 10 models in semi logarithmic scale.

Supplementary Note 7. Extending PQ-Net: Retrieving Site Occupancy Factor (sof)

Within the paper we have focused on the extraction of scale factors, lattice parameters and crystallite sizes which in the exemplar are sufficient to describe and understand the materials and process performance under operational conditions, and from our experience is that which is most commonly desired from XRD-CT and indeed majority of multiphase *in situ* experiments. In principle though the PQ-Net can be extended to other parameters and to illustrate, we demonstrate its use for extracting occupancy where the architecture is illustrated in Supplementary Figure 14 for a single-phase.

Batch size: 32

Training time: 2 ms/step

NiPd scale factor range: random sampling between $0-4 \times 10^{-9}$

NiPd lattice parameter range: $3.6 \pm 0.2 \text{ \AA}$

NiPd crystallite size a range: 2-25 nm

Ni1 site occupancy factor: random sampling between 0-1

Validation size: 10,000

Early stopping: 3

Added features: Poisson noise and 2nd degree Chebyshev polynomial

The NiPd lattice parameter range is given a much larger range than that introduced in the previous libraries to reflect real world lattice parameter changes that would accompany the occupancy ranges introduced. As previously, all parameters are normalised to 0-100 based on their sampling range while training. The errors before and after normalization are shown in Supplementary Table 14. The results validate that site occupancy can be extracted in this way.

Supplementary Table 14.

Parameter Name	Validation MAE (normalised)	MAE on original unit
Scale factor	1.62	6.48×10^{-9}
Crystallite size	2.54	$0.0102 \text{ \AA}$
Lattice parameter a	1.97	0.453 nm
site occupancy factor	2.46	0.0246